MAGS: Max-Gap Loss-Guided Siamese-Reconstruction Network for Hyperspectral Image Partial Label Learning

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***Abstract*— Due to the powerful feature extraction capabilities of deep learning, a series of deep learning-based methods for hyperspectral image (HSI) classification have been proposed and achieved satisfactory performance. However, most of these methods require a large number of labeled data, and the collection of completely accurate pixel-level labeled HSI data is difficult, resulting from the intricate label ambiguity of HSI and incomplete prior knowledge of annotators. Simultaneously, a few researchers focus on label ambiguity for HSI classification. Partial label learning (PLL) is one of the strategies to solve the problem where each training instance is assigned a candidate label set, among which only one is the ground truth label, which can essentially alleviate labeling difficulties. In this article, a max-gap loss-guided Siamese reconstruction network (MAGS) is proposed to combine PLL with HSI classification. MAGS consists of three components, including a spatial–spectral encoder, a spatial–spectral decoder, and a Siamese spatial–spectral encoder for high-quality feature representation learning to facilitate label disambiguation. In the encoding process, MAGS introduces the cross-attention and max-matching fusion strategies to obtain more representative features. In addition, to improve label disambiguation, the maximum gap loss is designed to guide the model training. Quantitative and qualitative results indicate that the MAGS outperforms several state-of-the-art methods on three HSI datasets. The code is available at https://github.com/Nemo96yu/MAGS.**

***Index Terms*— Hyperspectral image (HSI) classification, label disambiguation, partial label learning (PLL), spatial–spectral information.**

I. Introduction

HYPERSPECTRAL imaging is an advantaged technology that can capture the information with rich spectral [1], [2]. The generating hyperspectral image (HSI) has the advantage of high spectral resolution and spatial information simultaneously, which enables the acquisition of almost continuous spectral characteristic curves of land-covers. Thus, the HSI has been widely applied and achieved success in various domains such as environment surveillance, military reconnaissance, and so on [3]. As the basic data processing technique for these applications, HSI classification is to identify the specific type of land-covers for each pixel in HSI.

The key factor affecting the effectiveness of HSI classification is how to capture discriminative information through extracting spatial and spectral information hidden in HSI. Deep learning has proven to be the satisfactory framework for feature extraction in computer vision [4], such as convolutional neural network (CNN), recurrent neural network (RNN), and transformer network. Thus, in recent years, a series of deep learning-based methods for HSI classification have been proposed, which can be divided into three categories: the spectral information-based models that only rely on spectral information extraction, the spatial information-based models that only rely on spatial information extraction, and the spectral–spatial information-based models that focus on both spectral information and spatial information extraction. For the spectral information-based models, [5] utilized 1D-CNN to classify HSIs directly in the spectral domain, and [6] extracted spectral information via RNN that treats the HSI as sequential data. For the spatial information-based models, [7] combined Gabor wavelets with CNN kernels to mainly extract spatial features, Liu et al. [8] proposed a hybrid-convolution-guided information filtering and enhancement strategy to enhance the spatial feature extraction, [9] introduced a fine-label HSI cube generation approach to enhance the diversity of the spatial information of HSI. Neither the spectral information-based models and the spatial information-based models can incorporate spatial and spectral information, which is proved to be essential to provide more valuable information for HSI classification [4]. For the spectral–spatial information-based models, [10] introduced a spectral–spatial feature tokenization transformer (SSFTT) for capturing spectral–spatial information and obtaining high-level semantic features, [11] introduced two processes of multistage and multipath to fully explore spectral and spatial features, Jiang et al. [12] proposed a spectral–spatial wave network to improve the extraction capability of spectral–spatial information hidden in HSI based on CNN, and [13] presented a multiscale-CNN module to capture

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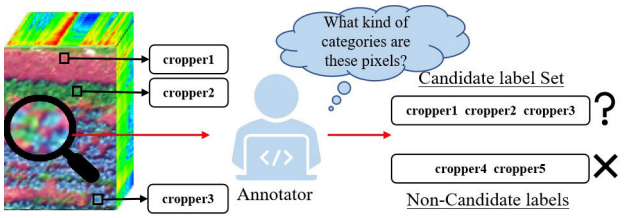


Fig. 1. Example for partial labeled data. In the magnified region, the boundaries between different features are not so distinct that it is difficult for the annotators to recognize the specific type of pixels in the region, and thus all types contained in the region are labeled to the pixels.

indicate spectral–spatial information from HSI. The above deep learning-based HSI classification methods have enhanced the capabilities for feature extraction and the improvement of HSI classification performance. However, to the best of our knowledge, most of the deep learning-based methods are essentially data-driven learning that requires a large number of labeled samples [14]. However, the collection of labeled samples is time-consuming and expensive, especially the pixel-level labeling of HSI data.

To reduce label pressure and label costs, some researchers tend to train deep learning models of HSI classification with small-size samples. Xu et al. [15] proposed a residual spatial attention kernel generation network for HSI classification when training samples are limited. Ma et al. [16] applied a semantic segmentation network to mitigate the small-size sample problem. Feng et al. [17] integrated assembling pixel pair features into the ensemble SMOTE to effectively handle small-sized HSI datasets. With the advent of few-shot learning, a mount of researchers introduced few-shot learning-based algorithms into the HSI classification [18], which enables learning from a small number of labeled samples. In [19], an adaptive subspace and feature transformation-based method is introduced to learn the HSI classifier with a few labeled samples. Then, a heterogeneous few-shot learning method was introduced in [20], which made full use of transferable knowledge from the natural image dataset and the spatial–spectral information. To further improve HSI classification performance, a few-shot learning framework based on the class covariance metric was proposed in [21].

Although such methods can alleviate the labeling cost to some extent, most of them ignore the fact that the inherent ambiguity of labels is the essential cause of labeling difficulties. In other words, the low interclass dissimilarity that existed in the vague area of the HSI would make it difficult for annotators to identify the correct class labels of the confused pixels. As shown in Fig. 1, there is a vague area in which the boundaries of different land-covers are fuzzy. Thus, the information could not support the annotator to determine which land-cover type the pixel is and only provides the scope of the categories. In this case, the annotator would like to assign the labels that seem likely to be the true label to the confused pixel, which makes up the candidate label set. This kind of data is called partially labeled data. Meanwhile, the emergence of the partial label learning (PLL) framework is designed to train the classification model for predicting the correct label of unseen instances with the partially labeled data. In other words, PLL is a kind of weakly supervised learning framework that allows training samples to be with the candidate label sets, among which only one ground-truth label is present and the others are false-positive labels [22].

The core of PLL is how to identify the ground-truth label from the candidate label set, namely, label disambiguation, which is used to provide effective supervised information for the training process. To achieve this, several PLL methods were proposed, which can be divided into the average-based disambiguation algorithm [23] and the identification-based disambiguation algorithm [24], [25]. The former refers to giving the same weight to each candidate label of each partial label sample and achieving disambiguation by combining the model's output on each candidate label, where the ground-truth output might be overwhelmed by the false-positive label. The latter refers to treating the ground-truth label of the partially labeled samples as hidden variables and optimizing the objective function of the embedded hidden variables in an iterative manner to achieve disambiguation, where the identified label might turn out to be the false-negative label. Furthermore, most of them focus the disambiguation strategies on the output space and ignore the important role that high-quality representation plays in label disambiguation.

To solve this problem, some deep learning-based methods have been proposed to focus on high-quality representation learning for high-quality label disambiguation, and they have achieved state-of-the-art results on various public datasets. In [26], researchers proposed a contrastive label disambiguation model [partial label learning with contrastive label disambiguation (PiCO)] that consists of a contrastive learning module in conjunction with a novel class prototype-based label disambiguation algorithm. In [27], researchers proposed a weighted loss of calibration rank to constrain the representation learning process, which achieved a satisfactory performance. In [28], a [partial-label learning with a guided prototypical classifier (PaPi)] is proposed, which can explicitly encourage representation to capture visual similarities between categories. However, most of these algorithms only consider the characteristics of natural images, and few of them take the characteristics of HSI into account. Recently, some researchers have made efforts to combine PLL with HSI classification, such as the superpixel-wise low-rank approximation (LRA)-based PLL method (SLAP) [29] and the unlabeled data-guided PLL (UPLL) [30], making it the pioneering solution and attracting the attention of researchers in the field of HSI classification.

Based on previous works, we further analyze the PLL framework by combining the characteristics of HSI. To the best of our knowledge, label disambiguation and representation learning are interdependent, and uncertainty in either one affects the other, which in turn can lead the model training process into a vicious circle. Although HSIs are rich in spatial and spectral information, the complexity of land covers (e.g., homogeneous and heterogeneous spectra) in HSI can make it more difficult for the alone encoding process to capture sufficient discriminative features for label disambiguation under limited supervision. The encoder–decoder structure has attracted a lot of attention due to its excellent performance in feature extraction [31], [32] for HSI classification,

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which is first proposed in unsupervised learning area, such as [33] and [34] introduced locality low-pass and preserving graph convolutional autoencoder, Lin et al. [35] proposed spatial–spectral masked autoencoder. Inspired by these methods, we think it might be a good manner to used it for a basic framework for HSI PLL. In the presence of false-positive labels in the candidate label set, the spatial–spectral features of HSI from the encoders might deviate from its discriminative features, which subsequently leads to the classifier cannot accurately identify the ground-truth label. Thus, enhancing the representation and robustness of feature extraction is important for PLL-based HSI classification. Recently, the Siamese network has been gradually applied to HSI classification, which lies in the efficient spectral feature extraction, the ability to adapt to high-dimensional HSI data, and handle the scarcity of labeled data [36], [37]. Obviously, the advantages of the Siamese network can be applied to HSI PLL to improve representation learning, which has not been fully utilized in HSI PLL. In addition, most PLL methods perform label disambiguation based on the predictions of the classifier, but they ignore the importance of matching the predictions of the classifier as much as possible with the ground truth to enhance the

To solve the above challenge, we proposed a max-gap loss-guided Siamese reconstruction network (MAGS) to extract the feature and promote label disambiguation. First, we adopt the encoder–decoder structure of the reconstruction network to ensure that the learned features can provide sufficient discriminative information for label disambiguation. In the process of feature encoding, we introduce the spatial–spectral interaction (SSI) encoders to extract the spatial–spectral information hidden in HSI, and use the SSI mechanism [38] to achieve the interaction of spectral and spatial features for capturing more valuable information. In addition, textual information about labels can provide extensive prior knowledge for HSI classification [39]. Thus, we further design the max-matching fusion strategy to achieve representation enhancement of feature extraction by fusing the semantic features of labels with the learned spatial–spectral representation. Second, to further enhance the robustness of the encoders in representation learning, we feed the noisy spatial–spectral inputs into the Siamese encoders that share the parameter weights, and establish alignment loss to achieve consistency between the Siamese encoders, namely, Siamese-contrastive encoder (SE). Finally, to enable the classifier's prediction results more match the ground-truth label and improve the capability of label disambiguation, the max-gap loss is designed to improve the model's ability to discriminate the ground-truth label from the candidate label set. The main contributions of this article are as follows.

1) To enhance feature representation, two simple yet effective modulars are integrated into spatial–spectral encoder. The SSI exploits the mutual characterization of spatial and spectral information which provides more valuable information for classification. The max-matching fusion ensures the enhancement of feature extraction. Consequently, both of them promote effective feature representation in different feature extraction processes, respectively.
2) In addition, an SE that shares parameter weights is first proposed to improve the robustness of representation learning for HSI PLL. We added noise to the spectral and spatial inputs and applied the alignment loss to achieve consistency between the Siamese encoders. Consequently, this Siamese spatial–spectral encoder can ensure robust representation learning across noisy inputs, which provides a new solution for overcoming the difficulty on feature extraction for HSI. PLL.
3) The max-gap loss is proposed to measure the gap between the predicted probabilities of the latent ground-truth and the negative labels, which should become larger and larger during the training process. This methodology ensures the enhancement of the model's ability to discriminate ground-truth labels. Consequently, the max-gap loss promotes and enhances the reliability of label disambiguation.

The organization of the rest of this article is as follows. The proposed MAGS method is described in Section II. Section III conducts some experiments on three public HSI datasets to demonstrate the effectiveness of MAGS. In Section IV, we analyze some conclusions of this article and suggestions for future work.

II. MAGS

HSI contains complex land-covers, and in many cases, it is difficult to label them into one certain category for some mixed land covers, which poses a great challenge to the labeling process. To alleviate this difficulty, some researchers have introduced the PLL framework, where each sample receives a set of labels with only one ground-truth label. Based on the PLL framework, we propose a MAGS to extract discriminative features of land-covers and facilitate label disambiguation. First, to capture sufficient discriminative features, we utilize the structure of encoder–decoder in the form of the reconstruction network, which consists of the SSI encoder and a spatial–spectral decoder. In the SSI encoder, we introduce the SSI for interaction between the spectral and spatial features and design the max matching fusion (MMF) module for acquiring more discriminative features. Second, to further enhance the robustness of feature representation, we construct the SE with the shared parameter weights of the spectral–spatial interaction encoder to process the noisy inputs and compute the alignment loss between the prediction results according to the feature representation of the above two encoders. Finally, to better match the prediction results of the classifier with the ground-truth label and achieve more reliable label disambiguation, we introduce the max-gap loss to make the gap between the max value and the remaining values in the prediction result as large as possible. The MAGS's architecture is shown in Fig. 2.

Given the original HSI data $\boldsymbol{I} \in \mathbb{R}^{m\times n\times l}$ with the label space $\mathcal{Y} = \{1, 2, \ldots, C\}$, $m \times n$ is the spatial size, l is the number of spectral bands and C is the number of class labels. For PLL in HSI, each pixel $I_i \in \boldsymbol{I}$ with l spectral bands has a set of

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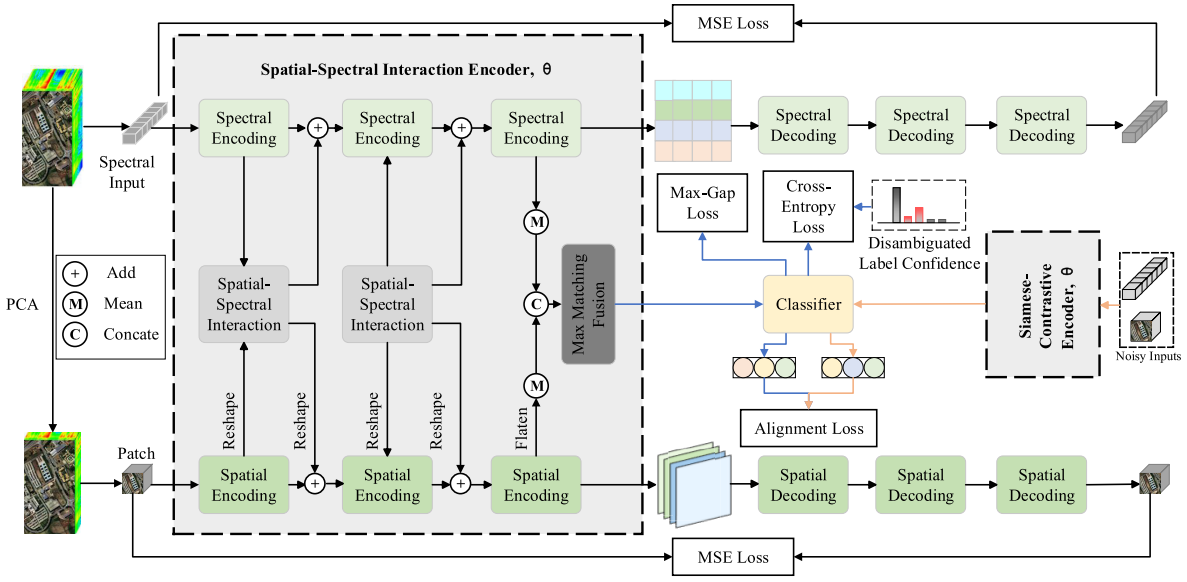


Fig. 2. Flowchart of the proposed MAGS method. Given the spectral input and spatial patch, the spatial–spectral encoder provides three types of output: one is the spectral feature representation for spectral encoding, one is comprehensive feature representation for the classifier, and one is spatial feature representation for spatial decoding. MAGS also leverages Siamese for noisy inputs to enhance the robustness of feature extraction, and introduces the max-gap loss for label disambiguation enhancing.

candidate labels $Y_i \in \mathcal{Y}$, of which only one label is the true label y_i. I_i denotes the spectral inputs and the data patch of each pixel is represented as X_i.

A. Spatial–Spectral Interaction Encoder

1) Spectral Encoding: Before the spectral encoding, we first map the spectral data of each pixel I_i into the feature map $I_i' \in \mathbb{R}^{l \times d_1}$ via one 1-D convolution layer with kernel size 1. Then, we split I_i' into l features vectors denoted as $\{s_1, s_2, \ldots, s_l\}$, $s_p \in R^{d_1}$ is the feature vector of pth spectrum in I_i corresponding to a time step. To obtain the long-range spectral information, we feed the sequence of l vectors into the long short-term memory (LSTM) as the spectral encoding layer whose output has the same dimension as its input. To be specific, we employ multiple LSTM layers with different hidden units to generate the spectral vector for each pixel along with the forward direction. At the time-step p, the hidden state can be updated with the aid of input and the $(p-1)$th step output

$$F_p' = f_{\mathrm{ReLU}}\left(f_{\mathrm{LSTM}}\left(F_{(p-1)}', s_p\right)\right) \tag{1}$$

where F_p' indicates the spectral semantic representation, $f_{\mathrm{ReLU}}(\cdot)$ is the activation function, and $f_{\mathrm{LSTM}}(\cdot)$ is the LSTM function. After reshaping the output vectors and arranging them into a map of size $l \times d_1$ to yield the spectral feature map $F' \in R^{l \times d_1}$. In this way, we can better capture long-dependencies information hidden in the spectral domain of each pixel via LSTM.

2) Spatial Encoding: We utilize the 2-D convolution layer to extract the hidden spatial information in HSI. We take the data patch X_i as input, and the detail process can be defined as follows:

$$F = f_{\mathrm{ReLU}}\left(f_{\mathrm{Pool}}\left(f_{\mathrm{Conv}}^{(k\times k, d_1)}(X_i)\right)\right) \tag{2}$$

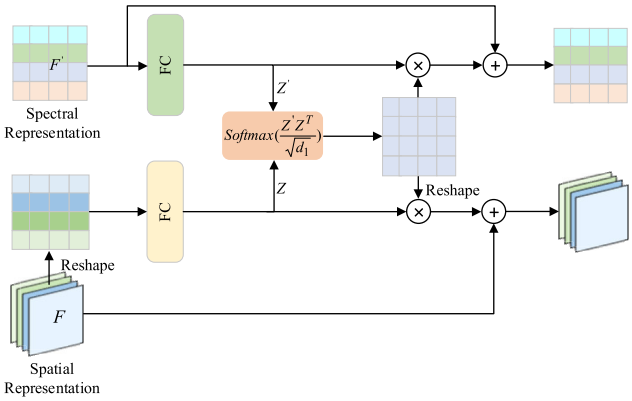


Fig. 3. Flowchart of SSI mechanism.

where $f_{\mathrm{Conv}}^{(k\times k, d_1)}(\cdot)$ indicates the 2-D convolution function with kernel size k and output dimension d_1, and $f_{\mathrm{Pool}}(\cdot)$ is the polling process. Convolution networks $f_{\mathrm{Conv}}^{(k\times k, d_1)}(\cdot)$ are used to capture spatial information F for each pixel.

3) SSI Mechanism: Most existing methods tend to extract spectral and spatial information separately during the encoding process and fuse them simply via concation or summation, which will waste a large amount of potentially valuable information in HSI. Thus, we design an SSI Mechanism to fit the characteristics of HSI for the interaction of spectral and spatial information. As shown in Fig. 4, given the spectral representation F' and the spatial representation F computed in (1) and (2), respectively, the whole process of the SSI mechanism is defined as follows:

$$Z' = f_{\mathrm{FC}}(F)$$
$$Z = f_{\mathrm{FC}}\left(f_{\mathrm{reshape}}(F')\right)$$

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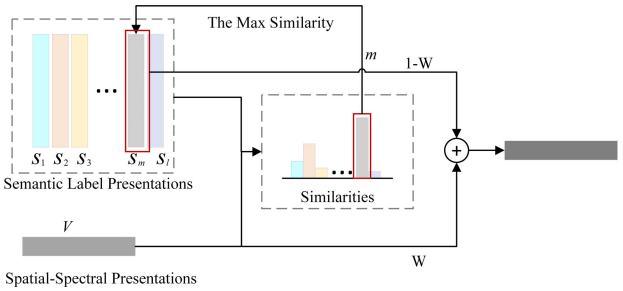


Fig. 4. Flowchart of MMF. After obtaining the feature representation from spatial–spectral encoders, MMF fuses them with the max similarity label's semantic representation to enhance features.

$$S = \text{softmax}\left(\frac{Z'Z^{\mathrm{T}}}{\sqrt{d_1}}\right)$$
$$F' = F' + \alpha(SZ)$$
$$F = F + \alpha(S^{\mathrm{T}}Z') \tag{3}$$

where $f_{\mathrm{FC}}(\cdot)$ indicates the fully connected function, $f_{\mathrm{reshape}}(\cdot)$ refers to changing the shape of the input to the desired shape, and α is the interaction weight between 0 and 1. As shown in (3), for Z' from the spectral encoding layer, we incorporate information by performing attention weighting S with Z from the spatial encoding layer. After that, we add attention values to F', F as inputs for the next spectral encoding layer and the spatial encoding layer, respectively.

4) Max Matching Fusion: After three layers of spatial and spectral encoding, we obtain the spectral representation $F' \in \mathbb{R}^{l\times d}$ and the spatial representation $F \in \mathbb{R}^{h'\times w'\times d}$, where d is the dimension of the representation, h' and w' are the size of the patch after convolution processing. As mentioned above, high-quality representation is very important to promote a virtuous cycle in HSI PLL. However, under the coarse supervised information provided by the candidate label set, it is difficult to acquire sufficiently distinguishable features from spectral and spatial information for label disambiguation. Semantic label representations generally serve as side information and have been shown to be effective in classification tasks [40]. Thus, in this section, we explore the semantic relation between semantic label representations and spatial–spectral features of HSI, which is expected to provide sufficient discriminative features for label disambiguation. Specifically, we pioneered the utilization of semantic label representations to further enhance discriminative features and introduce the MMF strategy that is shown in Fig. 4. First, we acquire a semantic label representation lookup table $S = \{S_1, S_2, S_3, \ldots, S_l\} \in \mathbb{R}^{l\times d'}$ by encoding the word of each label with a pretrained language model [e.g., bidirectional encoder representations from transformers (BERT)], where S_i denotes the semantic representation vector of ith label. Second, we process Spe and Spa in a 1-D space and concatenate them to obtain the spatial–spectral representation V as follows:

$$V_1 = f_{\mathrm{mean}}(F', \mathrm{dim} = 0)$$
$$V_2 = f_{\mathrm{mean}}(f_{\mathrm{Flatten}}(F), \mathrm{dim} = 0)$$
$$V = [V_1, V_2] \tag{4}$$

where $f_{\mathrm{Flatten}}(\cdot)$ is to flatten the 2-D spatial space to 1-D space.

Finally, we compute the similarity between the spatial–spectral representation V and the semantic representations of the labels and select the semantic representation of the most related label as a potentially useful feature, which can be added to the spatial–spectral representation as follows:

$$M = \left(f_{\mathrm{FC}}(S)V^T\right)$$
$$m = \mathrm{argmax}(M)$$
$$V = w \times V + (1 - w) \times S_m \tag{5}$$

where f_{FC} means to change the dimension of label semantic representation d' into the same dimension of V, and $\mathrm{argmax}(\cdot)$ indicates the process of select the maximum value. In this way, we can obtain a high-quality representation that will make it easier for the classifier to recognize the potentially true label for label disambiguation. Moreover, this part is just designed for the training process, and in the testing process, we directly feed the V computed from (4) into the classifier for label prediction.

B. Spectral–Spatial Decoder

As a kind of weakly supervised learning method, the PLL encoding process would be affected by false positive labels in the set of candidate labels. Therefore, we design the spectral encoding layer and the spatial encoding layer based on the reconstructed network to further enhance the reliability of feature representation via learning in a self-supervised manner. Specifically, given the spectral representation and spatial representation F', F of the spatial–spectral encoder, we design three spectral encoding layers and spatial encoding layers to restore them to the original input, respectively, as follows:

$$\tilde{I}_i = f_{\mathrm{Tanh}}\left(f_{\mathrm{LSTM}}\left(f_{\mathrm{ReLU}(f_{\mathrm{LSTM}}(f_{\mathrm{ReLU}}(f_{\mathrm{LSTM}}(F'))))}\right)\right)$$
$$\tilde{X}_i = f_{\mathrm{Tanh}}(f_{\mathrm{Conv}}(f_{\mathrm{ReLU}}(f_{\mathrm{Conv}(f_{\mathrm{ReLU}}(f_{\mathrm{Conv}}(F)))})) \tag{6}$$

where $\mathrm{Tanh}(\cdot)$ denotes the activation function of the last decoding layer. Then, we compute the mean squared error (mse) as the loss of consistency constraints to guide the reconstruction network training as follows:

$$l_{\mathrm{mse}} = \frac{1}{l}\sum(\tilde{I}_i - I_i)^2 + \frac{1}{hw}\sum(\tilde{X}_i - X_i)^2 \tag{7}$$

where l_{mse} indicates the per-pixel mse loss. In this way, we mitigate the negative impact of false positive labels in candidate labels set on the feature learning by aligning the decoded features to the original features.

C. Siamese-Contrastive Encoder

The Siamese network can extract information beyond labels from itself, thereby providing more stable and reliable information for HSI PLL. To further improve the robustness of the SSI encoder in terms of the feature representation capability, we add weighted random Gaussian noise to the spectral input and the patch input and feed them into the SE to obtain the spatial–spectral representation $\hat{V}$ via (4). Then, we feed V, $\hat{V}$ into the classifier to predict the label probability vector p, $\hat{p}$ of per-pixel. Based on this, we implement the per-pixel alignment loss term by minimizing the Kullback–Leibler (KL) divergence

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of the comprehensive label probability vector q and the label probability vector p and $\hat{p}$ as follows:

$$q = (p + \hat{p})/2$$
$$l_{\text{alig}} = D_{\text{KL}}(p||q) + D_{\text{KL}}(\hat{p}||q) \tag{8}$$

where q is the comprehensive label probability vector, and we, respectively, utilize the label probability vector p and $\hat{p}$ to guide q for achieving the overall consistency alignment, which will enhance the robustness of the model.

D. Label Disambiguation

As mentioned above, label disambiguation is a key step in combating the ambiguity of the PLL label. Like in previous works, we initialize the confidence vector of the label $\hat{Y}_i \in [0, 1]^C$ per pixel by normalizing its candidate label vector Y_i, where each entry denotes the probability that the corresponding label is the ground-truth label. Label disambiguation refers to updating $\hat{Y}_i$ progressively according to the probability vector $p = p_1, p_2, \ldots, p_l$ about the labels from the classifier, in which p_i is the prediction probability of ith label. To achieve this goal, we perform label disambiguation as follows:

$$\hat{Y}_{ij} = \begin{cases} \dfrac{p_i}{\sum_{k=1}^{C}(p_k)}, & \text{if } Y_{ij} \neq 0 \\ 0, & \text{if } Y_{ij} = 0. \end{cases} \tag{9}$$

In this way, more weights flow to more possible candidate labels slightly and progressively, while the weights of noncandidate labels are always zero. At the same time, $\hat{Y}_i$ serves as coarse supervision to calculate the classification loss (e.g., cross-entropy loss). Simultaneously, to ensure that the MMF module can select the most related label' semantic presentation, we also compute the cross-entropy loss between similarities M in (5) and $\hat{Y}_i$, called the semantic loss. The per-pixel classification loss is shown as follows:

$$l_{\text{cls}} = \sum_{j=1}^{C}(-\hat{Y}_{ij} \log p_j) + \beta \sum_{j=1}^{C}(-\hat{Y}_{ij} \log M_j) \tag{10}$$

where β is the weight of the semantic loss.

E. Max-Gap Loss

In essence, the process of label disambiguation aims to gradually increase the confidence of the true labels in the candidate label set while decreasing the confidence of the nontrue labels, and label disambiguation needs to be performed based on the output of the classifier. Based on the above findings, we believe that the gap between the predicted probability of the true labels and the predicted probability of the other nontrue labels in the output of the classifier needs to be as large as possible, and then we design the max-gap loss to measure to achieve this aim as follows:

$$p_m = \text{argmax}(p)$$
$$l_{\max} = \sum_{j \neq m}^{C}(e^{-(p_m - p_j)} - e^{-1}) \tag{11}$$

TABLE I
CATEGORY INFORMATION OF THE INDIAN PINES DATASET

Class	Color	Class Name	Training	Testing	Total
1		Alfalfa	2	44	46
2		Corn-notill	71	1357	1428
3		Corn-mintill	41	789	830
4		Corn	12	225	237
5		Grass-pasture	24	459	483
6		Grass-trees	37	693	730
7		Grass-pasture-mowed	1	27	28
8		Hay-windrowed	24	454	478
9		Oats	1	19	20
10		Soybean-notill	49	923	972
11		Soybean-mintill	123	2332	2455
12		Soybean-clean	30	563	593
13		Wheat	10	195	205
14		Woods	63	1202	1265
15		Building-Grass-Trees-Drives	19	367	386
16		Stone-Steel-Towers	5	88	93
		Total	512	9737	10249

where e^{-1} is the constant term to lower the loss to 0. In this way, we can obtain a more reliable rationale for label disambiguation and enhance the process of label disambiguation, which can promote a virtuous cycle in PLL. Then, we obtain the overall loss to update the network as follows:

$$\text{loss} = l_{\text{cls}} + w_1 l_{\text{mse}} + w_2 l_{\max} + l_{\text{alig}} \tag{12}$$

where w_1 and w_2 denote the weight between 0 and 1.

III. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we will verify MAGS on three public hyperspectral datasets and compare them with some of the state-of-the-art PLL and HSI classification algorithms.

A. Datasets

1) The Indian Pines dataset is an HSI of a region with Indian Pine trees in Indiana, USA, which has a size of 145 × 145 pixels with 200 spectral bands ranging from 0.4 to 0.25 μm. The detail information of this dataset is shown in Table I.
2) The PaviaU dataset is an HSI dataset over the city of Pavia, Italy, which consists of 610 × 340 pixels with 115 spectral bands. The detail of this dataset is shown in Table II.
3) The HoustonU dataset is an HSI of the University of Houston with its surrounding urban areas. The size of this image is 349 × 1905, which has 144 spectral bands and includes 15 types of land cover. The detail information is shown in Table III.

B. Data Processing

Given the original HSI data $\boldsymbol{I} \in \mathbb{R}^{m \times n \times l}$ with the label space $\mathcal{Y} = \{1, 2, \ldots, C\}$, $m \times n$ is the spatial size, l is the number of spectral bands and C is the number of class labels. For PLL in HSI, each pixel $I_i \in \boldsymbol{I}$ with l spectral bands has a set of candidate labels $Y_i \in \mathcal{Y}$, of which only one label is

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TABLE II
CATEGORY INFORMATION OF THE PAVIAU DATASET

Class	Color	Class Name	Training	Testing	Total
1		Asphalt	66	6565	6631
2		Meadows	186	18463	18649
3		Gravel	21	2078	2099
4		Trees	31	3033	3064
5		Metal sheets	13	1332	1345
6		Bare soil	50	4979	5029
7		Bitumen	13	1317	1330
8		Bricks	37	3645	3682
9		Shadows	10	937	947
		Total	427	42349	42776

TABLE III
CATEGORY INFORMATION OF THE HOUSTONU DATASET

Class	Color	Class Name	Training	Testing	Total
1		Healthy grass	63	1188	1251
2		Stressed grass	63	1191	1254
3		Synthetic grass	35	662	697
4		Tree	62	1182	1244
5		Soil	62	1180	1242
6		Water	16	309	325
7		Residential	63	1205	1268
8		Commercial	62	1182	1244
9		Road	63	1189	1252
10		Highway	61	1166	1227
11		Railway	62	1173	1235
12		Parking Lot 1	62	1171	1233
13		Parking Lot 2	23	446	469
14		Tennis Court	21	407	428
15		Runing Track	33	627	660
		Total	751	14278	15029

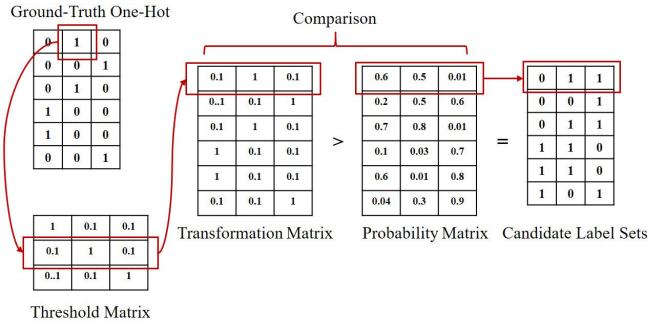


Fig. 5. Generation of the candidate label: construct uniform candidate labels by flipping each incorrect label with a probability $R = 0.1$.

the true label y_i. Flowing [26], we generate the candidate labels by transforming each negative label with probability $R(R < 1)$. Specifically, as shown in Fig. 5, where R is set to 0.1, we first construct the threshold matrix with a diagonal value of 1 and the remaining values of 0.1, and then locate the corresponding row of the threshold matrix according to the position of the value of 1 in the one-hot matrix. In this way, we can obtain the transformation matrix, where the number of rows is the number of samples, the number of columns is the number of class labels, and the column i of each row with a value of 1 represents its ground-truth label, to ensure that the ground-truth label of each sample is in the candidate label set. Finally, we generate a probability matrix of the same shape as the transformation matrix and uniformly distributed in the range of 0 to 1, and compare it with the corresponding value in the transformation matrix to obtain the candidate label set Y_i for each pixel (if it is less than the threshold, it is considered as a candidate label and its value is set to be 1, otherwise it is a noncandidate label and set to 0). In addition to taking the original pixels with spectral dimension l as input to the model, we also use the PCA to reduce the number of bands from l to b to obtain computationally friendly HSI data, from which we can create the data cube with patch size $h \times w$ for each pixel X_i to extract spatial information. The goal of PLL for HSI is to take advantage of the given data representation (e.g., I_i and X_i) to train the network under limited supervision (e.g., Y_i), which can predict the ground-truth labels of unseen pixels.

C. Experimental Setup

In the experiment, the labeled samples were divided by a certain percentage for training, and the rest samples were used for testing. The training samples were associated with the candidate label set and learned by using the MAGS algorithm to obtain the encoder and classifier, and then the test samples were embedded for semantic representation by the encoder and predicted for the true label by the classifier. We used the average accuracy of all classes (AA), the overall classification accuracy (OA), and the kappa coefficient (KC) as evaluation indicators.

In this article, we compare MAGS with several of the latest PLL algorithms. Traditional machine learning-based PLL methods include the following: SLAP [29], PL-AGGD [24], and DPCLS [25]. The deep learning-based PLL methods include: PiCO [26], PaPi [28], and ParSE [27]. To further validate the effectiveness of representation capability, we selected several fully supervised algorithms including: LCMR [41], SSFTT [10], and Gnet [11] for comparison when R is set to 0.

In the experiment, we randomly selected 5%, 1%, and 5% samples from each class in the Indian Pines, PaviaU, and HoustonU datasets for training, respectively, and the rest as test samples. The verification experiments of the proposed method were all implemented in the Pytorch environment, using an NVIDIA GeForce 3090Ti 24-GB GPU. The Adam optimizer was selected as the initial optimizer, and the original learning rate was set to 0.001. For batch training, the size of each minibatch was set to 128. The training epoch was set to 100 for Indian Pines, PaviaU, and HoustonU, respectively.

D. Main Results

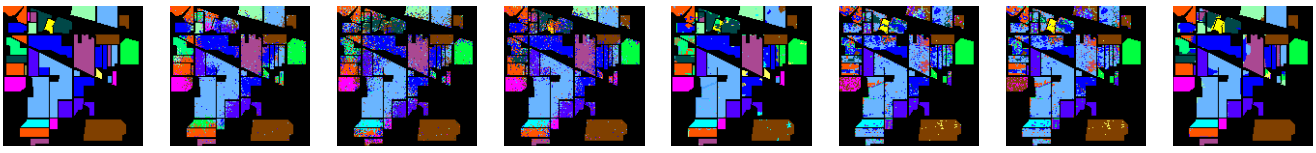
The quantitative results of different methods with $R = 0.3$ are reported in Table IV for Indian Pines, PaviaU, and HoustonU, respectively, and the optimal results are denoted in bold. Among traditional machine learning-based PLL methods, DPCLS and PL-AGGD are much worse than SLAP, which is due to the fact that they originally designed the algorithm based on natural images and did not take the characteristics of HSI into account. Among the performance of deep learning-based PLL methods, PiCO, PaPi, and ParSE can obtain performance better than DPCLS and PL-AGGD in

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TABLE IV

CLASSIFICATION ACCURACY (EACH CLASS ACCURACY, AA, OA, AND KC) ON INDIAN PINES, PAVIAU, AND HOUSTONU (%) WITH $R = 0.3$

Dataset	class	SLAP	PL-AGGD	PiCO	PaPi	ParSE	DPCLS	MAGS
Indian Pines	Alfalfa	**100**	0.00	0.00	0.00	0.00	0.00	54.55
	Corn-notill	87.68	52.51	81.43	63.82	78.92	46.83	**94.77**
	Corn-mintill	82.23	32.23	84.66	55.89	71.74	36.17	**92.90**
	Corn	65.78	11.11	**89.82**	41.15	0.00	12.89	72.00
	Grass-pasture	**96.07**	39.30	54.90	57.73	69.50	13.76	84.31
	Grass-trees	97.26	97.69	86.74	65.71	73.63	97.55	**99.57**
	Grass-pasture-mowed	**76.92**	0.00	0.00	0.00	0.00	0.00	0.00
	Hay-windrowed	96.70	100.00	93.41	87.69	90.33	100.00	**100.00**
	Oats	31.58	0.00	0.00	0.00	0.00	0.00	0.00
	Soybean-notill	**87.65**	14.30	78.03	60.28	72.40	10.83	82.88
	Soybean-mintill	87.44	81.00	**93.01**	83.63	88.94	82.85	**97.81**
	Soybean-clean	89.34	24.87	0.00	44.50	56.38	15.99	**84.37**
	Wheat	99.48	92.27	0.00	73.85	71.79	87.63	**100.00**
	Woods	97.59	95.42	96.76	96.51	94.26	96.17	**99.83**
	Buildings-Grass-Trees-Drives	96.45	12.84	88.28	52.59	59.67	5.19	**95.10**
	Stone-Steel-Towers	36.36	82.95	0.00	0.00	24.72	88.64	**100.00**
	AA	**83.03**	46.03	52.94	48.96	53.27	43.41	78.63
	OA	89.51	60.73	78.30	70.11	76.50	58.44	**93.34**
	KC	88.12	54.06	75.08	65.02	72.96	51.21	**92.40**
Dataset	class	SLAP	PL-AGGD	PiCO	PaPi	ParSE	DPCLS	MAGS
PaviaU	Asphalt	84.48	87.19	79.97	83.32	78.29	88.10	**97.01**
	Meadows	97.99	99.52	75.31	87.66	94.07	99.63	**100.00**
	Gravel	**88.40**	24.93	28.19	33.86	33.96	14.39	72.04
	Trees	48.73	69.07	46.57	3.10	47.13	66.80	**81.24**
	Metal sheets	95.19	98.50	0.00	20.35	26.95	98.27	**99.92**
	Bare Soil	75.65	31.76	49.97	57.04	68.57	23.12	**95.68**
	Bitumen	94.30	0.00	32.57	60.74	29.69	0.00	**99.32**
	Bricks	84.77	87.98	71.97	63.03	64.67	90.10	**91.96**
	Shadows	81.64	**99.04**	0.00	14.82	0.00	98.93	98.40
	AA	83.46	66.44	42.73	47.10	49.26	64.37	**92.84**
	OA	87.57	79.67	63.03	68.02	73.59	78.34	**95.56**
	KC	83.56	71.64	52.42	56.77	63.40	69.60	**94.07**
Dataset	class		PL-AGGD	PiCO	PaPi	ParSE	DPCLS	MAGS
HoustonU	Healthy grass		83.16	84.95	78.22	73.09	83.08	**93.52**
	Stressed grass		81.44	43.96	58.39	59.14	79.09	**97.31**
	Synthetic grass		99.70	92.91	90.20	91.70	100.00	**96.22**
	Trees		98.81	78.93	76.99	75.38	**99.24**	99.15
	Soil		99.75	92.63	90.85	90.42	99.75	**100.00**
	Water		84.42	5.50	47.25	50.81	84.74	**89.97**
	Residential		90.53	77.18	65.15	69.29	89.29	**91.12**
	Commercial		50.89	40.02	43.91	42.05	48.35	**87.14**
	Road		82.51	70.34	62.27	69.58	86.04	**93.19**
	Highway		51.33	92.11	95.03	93.65	39.06	**99.91**
	Railway		44.76	81.69	80.15	84.41	33.16	**96.68**
	Parking Lot 1		18.36	76.19	76.71	75.09	17.16	**99.40**
	Parking Lot 2		1.80	0.00	65.02	56.28	2.02	**87.89**
	Tennis Court		68.97	0.00	94.35	83.29	68.23	**100.00**
	Running Track		99.84	89.15	90.91	88.52	99.68	**100.00**
	AA		70.42	61.70	74.36	73.51	68.59	**95.43**
	OA		71.09	69.55	74.11	73.96	68.85	**95.69**
	KC		68.68	66.82	72.03	71.84	66.25	**95.34**

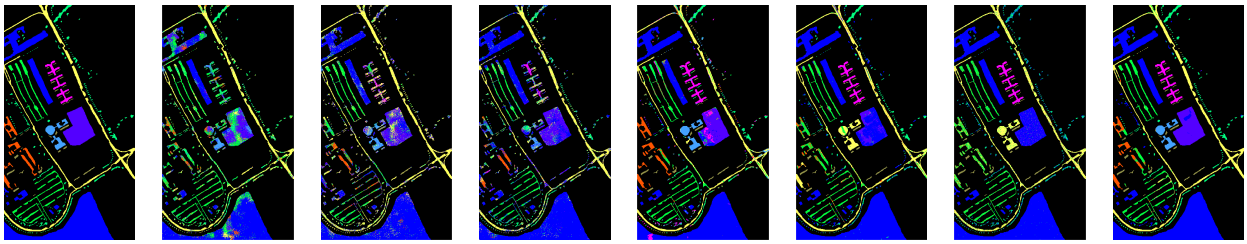


(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Classification maps on the PaviaU dataset with $R = 0.1$. (a) Ground-truth map. (b) PiCO. (c) PaPi. (d) ParSE. (e) SLAP. (f) PL-AGGD. (g) DPCLS. (h) MAGS.

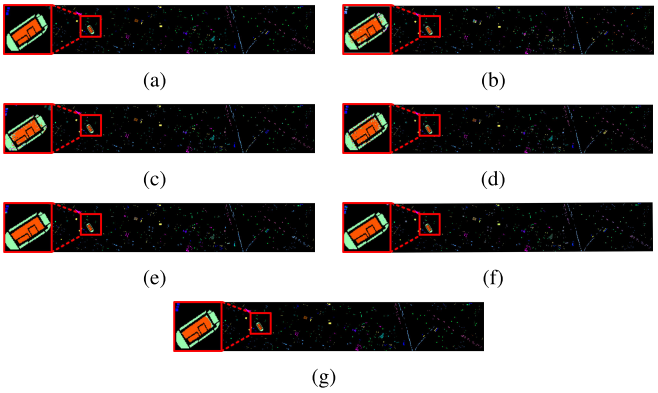
most cases on Indian pines and HoustonU dataset, but they are still relatively poor compared with SLAP, which is the well-designed algorithm to combine HSI classification and PLL. Notably, the performance of PiCO, PaPi, and ParSE

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Classification maps on the PaviaU dataset with $R = 0.1$. (a) Ground-truth map. (b) PiCO. (c) PaPi. (d) ParSE. (e) SLAP. (f) PL-AGGD. (g) DPCLS. (h) MAGS.



(a) (b) (c) (d) (e) (f) (g)

Fig. 8. Classification maps on the PaviaU dataset with $R = 0.1$. (a) Ground-truth map. (b) PiCO. (c) PaPi. (d) ParSE. (e) PL-AGGD. (f) DPCLS. (g) MAGS.

TABLE V
CLASSIFICATION ACCURACY ON INDIAN PINES DATASET WITH DIFFERENT VALUE OF R

Models	R=0.1		R=0.2		R=0.3	
	OA	KC	OA	KC	OA	KC
SLAP	92.67	91.65	91.06	89.84	89.51	88.12
PL-AGGD	64.98	59.17	61.59	54.81	60.73	54.06
PiCO	85.84	83.78	80.50	77.72	78.30	75.08
PaPi	80.78	77.68	77.34	73.80	70.11	65.02
ParSE	83.97	81.62	78.05	74.71	76.50	72.96
DPCLS	63.93	58.05	61.03	54.18	58.44	51.21
MAGS	**96.28**	**95.75**	**94.71**	**93.96**	**93.34**	**92.40**

Fully Supervised (R=0)

Models	OA	KC
LCMR	95.87	95.29
SSFTT	94.55	93.78
Gnet	95.07	94.37
MAGS	**96.41**	**95.90**

on the PaviaU dataset is worse than DPCLS and PL-AGGD, which denotes that the performance of deep learning-based PLL algorithms on HSI data is worse than traditional machine learning-based PLL methods when the training sample ration is as low as 1% on PaviaU dataset. The evaluation data clearly show that the proposed method achieves the best performance in terms of OA, AA, and KC in most cases. Specifically, the experimental results demonstrate good generalization of the model owing to three points: the SSI encoder extracts sufficiently discriminative features, the encoding-decoding structure enhances the robustness of the feature representation and the the max-gap loss facilitates the process of label disambiguation. In addition, models except for SLAP perform poorly in "Grass-pasture" and "Oats" of the Indian Pines dataset, which are the unbalanced classification results. Looking insight into Table I, the number of training samples of these categories is only one, which results in sample imbalance and is not conducive to models' training of deep learning-based methods. By further observation, it is easy to see that among the methods that do not take the characters of HSI data into account, machine learning-based methods (including PL-AGGD and DPCLS) perform better than deep learning-based methods (including PiCO, PaPi, and ParSE) on the categories "water" of HoustonU dataset and the categories "Shadows" of Indian Pines dataset with smallest number of training samples, which denotes the former may be more advantageous in the face of sample imbalance. Moreover, it should be noted that MAGS performs worse than SLAP in terms of AA on the Indian Pines dataset, which can be inferred that when the training samples of the categories are imbalanced, it is difficult for the deep learning-based model to learn effective features for the categories with a very small number of training samples. Thus, the deep learning-based model cannot achieve satisfactory performance in these categories, which leads to worse accuracy on AA.

The classification maps of several comparison models are shown in Figs. 6–8 for the Indian Pines, PaviaU, and HoustonU datasets, respectively. By visual analyzing, MAGS obtains the cleanest and closet maps to the ground-truth maps. The comparison models except for SLAP obtain the classification maps that contain significant noise for the three datasets, which indicates that these methods cannot accurately identify the types of land-covers. In addition, SLAP obtains relatively clean maps but performs poorly compared to MAGS. For the Indian Pines dataset, the comparison shows that the big brown square in the lower right of each image in Fig. 6 is a relatively noisy area, and the proposed models identifies the brown areas to a greater extent. For the PaviaU dataset, the category "Bare soil" in the center area in Fig. 7 is identified clearest by MAGS. In addition, for the HoustonU dataset, by comparing the brown area shown in Fig. 8, MAGS still maintains the best performance. All in all, the visual evaluation demonstrates the superiority of MAGS.

To further validate the effectiveness of label disambiguation, we set R to 0.1, 0.2, and 0.3 and show the results in Tables V–VII. It is obvious that the proposed method exhibits the stability in classification performance and achieves the

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TABLE VI
CLASSIFICATION ACCURACY ON PAVIAU DATASET WITH DIFFERENT VALUE OF R

Models	R=0.1		R=0.2		R=0.3	
	OA	KC	OA	KC	OA	KC
SLAP	93.29	91.10	90.32	87.22	87.57	83.56
PL-AGGD	81.94	75.18	80.33	72.96	79.67	71.64
PiCO	79.99	73.52	76.85	69.29	63.03	52.42
PaPi	77.95	70.68	73.96	65.05	68.02	56.77
ParSE	80.62	73.90	76.05	67.20	73.59	63.40
DPCLS	81.44	74.44	78.92	70.98	78.34	69.60
MAGS	**96.68**	**95.58**	**95.68**	**94.22**	**95.56**	**94.07**

Fully Supervised (R=0)

Models	OA	KC
LCMR	**97.45**	**96.62**
SSFTT	95.64	94.20
Gnet	94.32	92.28
MAGS	97.22	96.30

TABLE VII
CLASSIFICATION ACCURACY ON HOUSTONU DATASET WITH DIFFERENT VALUE OF R

Models	R=0.1		R=0.2		R=0.3	
	OA	KC	OA	KC	OA	KC
PL-AGGD	72.56	70.28	71.10	68.70	71.09	68.68
PiCO	85.41	84.16	80.71	79.11	69.55	66.82
PaPi	80.93	79.39	78.56	76.82	74.11	72.03
ParSE	82.50	81.08	79.85	78.22	73.96	71.84
DPCLS	72.36	70.07	69.79	67.28	68.85	66.25
MAGS	**97.73**	**97.55**	**97.42**	**97.21**	**95.69**	**95.34**

Fully Supervised (R=0)

Models	OA	KC
LCMR	93.09	92.53
SSFTT	95.41	95.03
Gnet	90.31	89.48
MAGS	**97.84**	**97.67**

SOTA results under various partial label rates, particularly excelling in scenarios with high levels of partial label rate. Specifically, MAGS outperforms the other models by more than 2% on the whole dataset under various values of R. It should also be noted that the superiority of the proposed model is more significant when the value of R is larger within a certain range, indicating the robustness of MAGS in the PLL HSI classification. Last but not least, we further validate the model performance under fully supervised conditions with three HSI classification models. It is obvious that the proposed model outperforms the other methods on the Indian Pines and HoustonU datasets, which can be inferred to be MAGS's superiority in representation learning. It is worth noting that Gnet's results are quoted from the original article and the number of training samples is larger than this article, but their classification accuracy is lower than that of MAGS.

E. Ablation Results

To fully demonstrate the effectiveness of the proposed method, ablation experiments were carried out according to the combination of different strategies on the Indian Pines, PaviaU, and HoustonU datasets with $R = 0.3$. The influence of four strategies was analyzed in terms of classification accuracy, and the results are shown in Table VIII. Specifically, the whole model includes four strategies: max-gap loss ($l_{\max}$), SE, SSI, and MMF. First, we validate the effectiveness of max-gap loss under two conditions: uniform pseudo target (without label disambiguation) and soft label disambiguation and it is obvious the model without max-gap loss yielded the worse classification accuracy. Looking at Fig. 10, we can also observe the clear downward trend on the value of the max-gap loss. In this way, it's easy to demonstrate the effectiveness of max-gap loss on prompting label disambiguation. Second, without the SE, the performance of the model drops significantly, which demonstrates that it plays a significant importance role in feature representation learning. Third, if the model removes the SSI module, its performance also drops to some extent, which indicates that the interaction between spatial and spectral features provides more valuable information for HSI classification. Finally, the MMF module is also necessary for the model by using semantic label representations to prompt the discriminative features for label disambiguation, which improves classification accuracy in most cases. In conclusion, the analysis of the ablation experimental results further confirms the validity of our model.

TABLE VIII
ABLATION RESULTS ON INDIAN PINES, PAVIAU, AND HOUSTONU WITH $R = 0.3$

Dataset	L_{max}	SE	SSI	MMF	OA	AA	KC
Indian Pines	Uniform Pseudo Target						
	–	√	√	√	68.28	54.32	64.13
	√	√	√	√	**86.74**	**71.97**	**84.80**
	Soft Disambiguation						
	–	√	√	√	91.96	73.99	90.78
	√	–	√	√	83.53	66.00	81.03
	√	√	–	√	90.43	72.27	89.01
	√	√	√	–	91.34	76.73	90.09
	√	√	√	√	**93.34**	**78.63**	**92.40**
PaviaU	Uniform Pseudo Target						
	–	√	√	√	93.02	86.67	90.67
	√	√	√	√	**95.14**	**91.81**	**93.50**
	Soft Disambiguation						
	–	√	√	√	95.37	91.52	93.80
	√	–	√	√	93.36	90.32	91.07
	√	√	–	√	95.28	92.09	93.69
	√	√	√	–	93.87	90.72	91.79
	√	√	√	√	**95.56**	**92.84**	**94.07**
HoustonU	Uniform Pseudo Target						
	–	√	√	√	51.25	52.03	47.42
	√	√	√	√	**73.51**	**70.95**	**71.37**
	Soft Disambiguation						
	–	√	√	√	96.21	96.09	95.90
	√	–	√	√	95.18	94.52	94.79
	√	√	–	√	94.31	93.77	93.84
	√	√	√	–	94.05	93.24	93.56
	√	√	√	√	**95.69**	**95.43**	**95.34**

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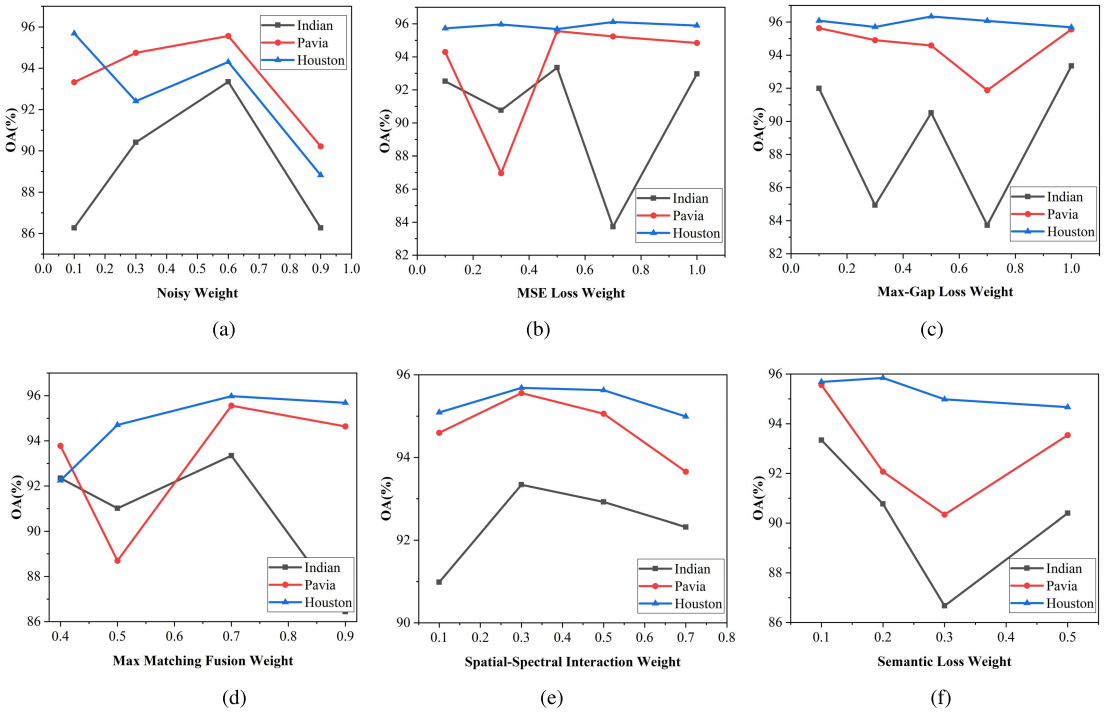


Fig. 9. Classification accuracy with respect to different values of parameters. (a) Noisy weight. (b) MSE loss weight w_1. (c) Max-gap loss weight w_2. (d) MMF weight w. (e) SSI weight α. (f) Semantic loss weight β.

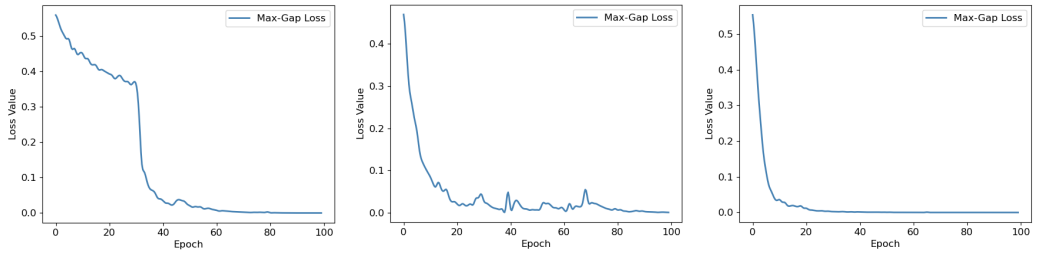


Fig. 10. Trend of max-gap loss with $R = 0.3$. (a) Indian Pines. (b) PaviaU. (c) HoustonU.

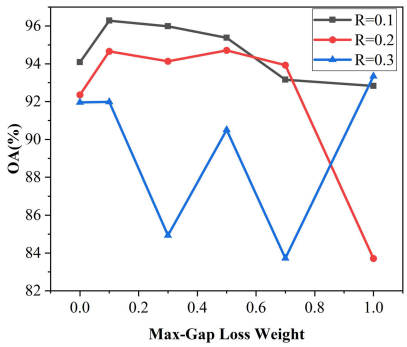


Fig. 11. Relation between partial label data and max-gap loss weight on Indian Pines Dataset.

F. Parameter Analysis

In MAGS, there are four important parameters including the noisy weight, mse loss weight, max-gap loss weight, and MMF weight. In this section, we analyzed the change of OAs under different parameter settings with $R = 0.3$ and listed the results in Fig. 9. First, by taking insight into Fig. 9(a), within a certain range, increasing the noise improves the classification accuracy on the Indian Pines and PaviaU datasets. Thus, to ensure the overall optimization of the model, we set the noise weight of the model as 0.6 on the Indian Pines and PaviaU datasets. Simultaneously, it's obvious that the HoustonU dataset is more sensitive to noise and it can achieve the satisfactory performance when the noisy weight is set to 0.1. Second, by observing from Fig. 9(b), we can find the model would achieve the optimal results when the weight of mse loss is set to 0.5 on the Indian Pines and PaviaU datasets, and the HoustonU dataset is not sensitive to the weight of mse loss. Third, it is easy to find out from Fig. 9(c) that the model achieves the best performance under the condition of $R = 0.3$ when the weight of max-gap loss is set to 1. And then, it's worthy to note from Fig. 9(e) and (f) that the MAGS can achieve the overall optimization when the SSI weight is set as 0.3 and the semantic loss weight is set to 0.1. Moreover, we further analyzed the relation between the weight of max-gap loss and the partial label threshold R on the Indian

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TABLE IX
TIME CONSUMPTION AND ROBUSTNESS ON INDIAN PINES DATASET

Model	PiCO	ParSE	PaPi	MAGS
training epochs	800	300	300	100
training Time(s)	795.23	330.53	400.84	28.56
testing Time(s)	12.99	17.68	16.26	2.87
Avg OA(%)	75.23	74.05	70.45	93.00
Std	3.98	1.41	0.58	0.69

dataset and listed the results in Fig. 11. It's noteworthy that the larger R is, the larger the weight of max-gap loss required for the optimal performance of the model is, indicating that the more serious the label ambiguity is, the more effective the positive guiding effect of maximizing loss on the label disambiguation process is. Therefore, in some cases, it is necessary to appropriately adjust the weight of max-gap loss according to the change of R to make the model achieve satisfactory results in some. Finally, by taking an insight into Fig. 9(d), the weight of MMF also serves difference influence on different datasets, and we set it 0.7, 0.7, and 0.9 for Indian pines, PaviaU, and HoustonU, respectively.

G. Time Complexity and Robustness Analysis

As shown in Table IX, we also conduct the experimental comparison for the time consumption and robustness on several deep learning-based comparison methods (including PiCO, ParSE, and PaPi) on the Indian Pines dataset. In this experiment, we observed significant disparities in the number of training iterations required by the different comparison models. For instance, the model proposed in our article requires only 100 training iterations, while the SOTA methods, such as ParSE and Parse, require 300 iterations, and PiCO requires as many as 800 iterations. In addition, these comparison models utilize ResNet architectures that are much larger than the ones used in our model, often involving significantly more parameters. As a result, the overall runtime of these models is considerably longer compared to the proposed approach in this article. In this way, we can find out the superiority of our model to compare deep learning models on time consumption, which is an important aspect of model evaluation in deep learning research. We conduct five independent experiments when R is 0.3, and compute the average OA, standard deviation (Std). In this way, we can find that our model still achieves the best performance on average OA, and show good stability through variance and standard deviation.

IV. CONCLUSION

In this article, we propose an HSI PLL method, termed the MAGS, to solve the problem of label ambiguity that results from the HSI labeling process. We first obtain the coarse spatial–spectral information based on the reconstruction network, which consists of the spatial–spectral encoder and spatial–spectral decoder. During the process of SSI encoding, we use the SSI machine to achieve the interaction between spatial information and spectral information, and introduce the MMF strategy to further extract the discriminative features from the spatial–spectral representation. In this way, we can obtain high-quality feature representations for label disambiguation. Furthermore, we add Gaussian noise to the inputs and introduce the SE to enhance the robustness and reliability of the feature representation. In addition, to reduce the influence of false-positive labels, we develop the max-gap loss to guide the process of label disambiguation and model training. Based on the above design, we find potentially true labels from the candidate label set and provide proper guidance for the model training. From the experimental results on the Indian pines, PaivaU, and HoustonU datasets, MAGS can effectively improve the classification performance of HSI PLL compared with other related algorithms. However, the model tends to ignore feature extraction for categories with fewer samples when the data are unbalanced, leading to poor classification accuracy regarding that category. In future work, we will try to solve this shortcoming and focus on the open-set classification [42].

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